



Study BOX

One Question. A Complete Study Kit.

Theme

Open Innovation

Core idea

Answer every question in text, visuals, video and slides.

Focus

Understanding, engagement & exam-readiness

Problem Statement

Finding an answer is easy. Understanding it is harder.

Everyday study

Search a topic

Open many tabs

Make notes

The friction

Text-only answers

Scattered resources

Hard to visualise

The central question

How can one question
give a complete, easy-to-
understand answer?

Question

>

Chat reply

>

Search videos

>

Find diagrams

>

Make slides

>

Lost time

Existing Solutions & The Gap

Search engines

Links and pages

Gap: you collect and sort everything yourself.

AI chatbots

Fast text answers

Gap: mostly text, casual in style.

Video platforms

Visual lessons

Gap: not tailored to your exact question.

Note and slide tools

Organise and present

Gap: you create the content yourself.

Identified gap

Most tools give one format. Study BOX delivers every format from one question.

Proposed Solution

Turn every question into a complete study kit.

The shift

Instead of: “Search, read, watch and make notes separately.”

Think: “Ask once and get everything.”

The principle

The real question stays the objective.

Study BOX adds visuals, video, slides and a quiz around the answer.

Question

›

Formal answer

›

Visuals & graphs

›

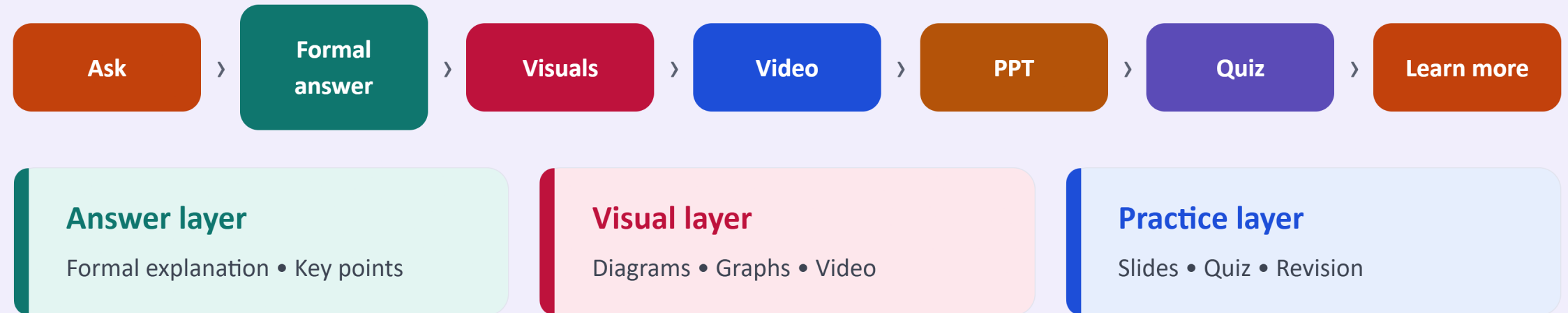
Video & PPT

›

Quiz

The Study Loop

Every answer ends with a way to check understanding.



Key Features

Formal answers

Textbook-style, structured explanations.

Visuals & graphs

Diagrams and charts for the topic.

Video lessons

A matching YouTube video.

Auto PPT

Slides generated from the answer.

Quiz

Check understanding instantly.

Interactive diagrams

Try the concept hands-on.

Typo-friendly

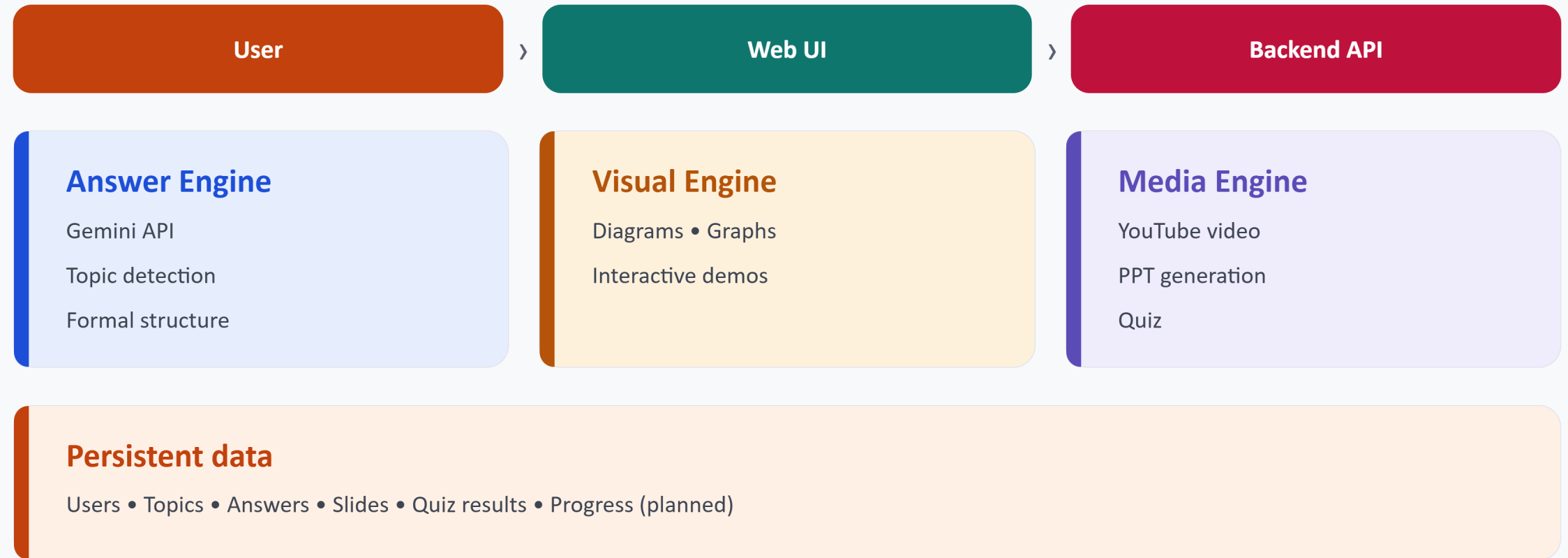
Understands misspelled questions.

Syllabus-aware (planned)

Answers tailored to your course.

Technical Approach

A full-stack system turns one question into a complete study kit.



Technology Stack

Frontend

HTML / CSS / JavaScript — User interface

Backend

Python + FastAPI — API layer

AI

Google Gemini API — Answers and slide content

Video

YouTube search — Matching video lessons

Output

PPT generation — Downloadable slides

Prototype

Live demo — Binary Search Tree walkthrough

Design principle

Problem → Requirement → Technology → Solution

Feasibility, Limitations & Impact

Feasibility

- Established tools (FastAPI, Gemini API)
- Working prototype for one topic
- Clear answer, visual and media parts
- Web first, mobile later

Limitations

- AI answers need accuracy checks
- Auto-made visuals vary in quality
- Video match depends on search results

Potential impact

- School and college students
- Teachers needing quick slides
- Self-learners
- Study groups

Core value

Change the study loop: from “search and sort” to one question that gives a complete answer.

The Core Idea

A different question for study tools.

Traditional approach

How can we get a better answer?

Question → Text reply

Study BOX approach

What if one question returned everything needed to understand it?

Question → Answer → Visual → Video → Slides → Quiz

Question

›

Answer

›

Visual

›

Video

›

Slides

›

Quiz



Thank You

Questions & Discussion

Judge discussion

Why this problem? • Why this approach? • What are the trade-offs?